

TEAM :
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IDENTIFYING THE OUTLIERS USING SPECTRAL CLUSTERING

Abstract: Outlier detection has a wide range of applications in many areas such as fraud detection, discovery of criminal activities in electronic commerce etc. Various techniques have been developed to detect outliers. In this paper, we discuss Spectral clustering, an algorithm to obtain the abnormal data as outliers by using the information of eigenvalues and eigenvectors statistically. The algorithm takes the number of neighbourhoods shared by the objects as the similarity measure to construct a spectral graph.

INTRODUCTION

Spectral clustering is a machine learning algorithm used for clustering data points based on their similarity. It has been widely used in various applications such as image segmentation, community detection in social networks, and document clustering. It is particularly useful when the data is not easily separable in the original feature space, but can be separated in a transformed space. One of the benefits of spectral clustering is that it can be used to identify outliers, which are data points that do not belong to any of the clusters.

PROBLEM STATEMENT

To identify malicious dealers in a network based on their transactions. We need to do this using spectral clustering.

DESCRIPTION OF THE DATASET

	cov1	cov2	cov3	cov4	cov5	cov6	cov7	sal_pur_rat	igst_its_tot_its_rat	lib_igst_its_rat
0	0.997797	0.999888	0.215934	0.196713	0.000000	0.955616	0.998810	-0.032581	1.761759	-0.054329
1	0.994004	0.979902	-0.337135	-0.248634	0.000000	0.640812	0.553918	-0.032026	-0.629311	-0.053516
2	0.947603	0.455667	0.001743	0.128610	-0.004054	-0.162069	0.960601	-0.030209	1.535697	-0.054215
3	0.396577	0.919933	0.496451	0.576824	-0.340718	0.802363	0.673710	-0.032058	0.449160	-0.054126
4	0.999893	0.327615	0.700477	0.315601	0.000000	0.300785	0.979009	-0.032224	1.762049	-0.054330

- The dataset is provided in the data.csv file.
- Each row corresponds to a dealer.
- The first seven columns are correlation parameters which lie in the range -1 to 1.
- The last three columns are ratio parameters.

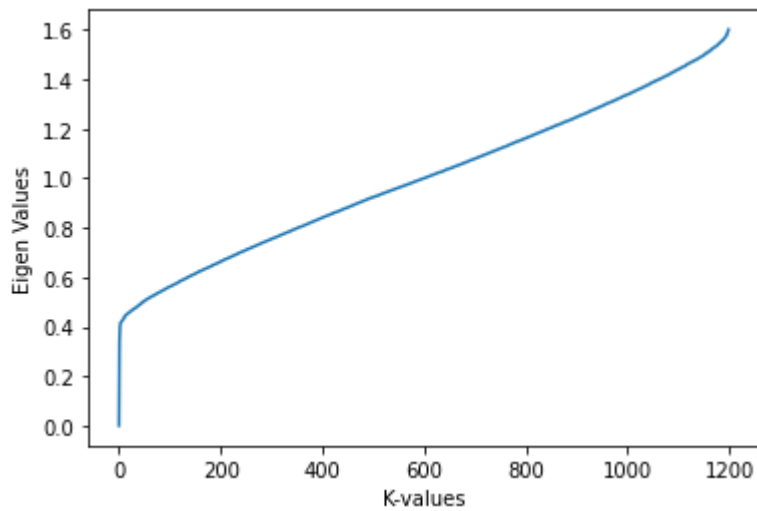
ALGORITHM USED : Spectral Clustering

The process to identify outliers can be broken down into the following steps:

1. Construct a similarity graph: The first step is to construct a similarity graph from the input data. The graph is constructed using the Gaussian kernel method.
2. Compute the Laplacian matrix: The Laplacian matrix is a matrix derived from the similarity graph. It measures the differences between neighbouring points in the graph.
3. Compute the eigenvectors and eigenvalues: The eigenvectors and eigenvalues of the Laplacian matrix are computed using matrix decomposition techniques such as Singular Value Decomposition (SVD).
4. Project data onto new feature space: The eigenvectors are used to transform the data into a new feature space, where the clusters can be easily identified.
5. Cluster data in the new feature space: Once the transformed data is obtained, k-means clustering is used to identify the clusters of dealers who exhibit similar transactional behaviour. Spectral clustering can identify the (potentially malicious dealers) outliers by grouping the other points into clusters, leaving the outliers as unassigned points.

RESULTS:

The plot shows that the graph is connected. But the significance difference between the 1st and 2nd eigenvalues shows that we can classify data into 2 clusters.



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Clusters: [1 0 1 ... 0 0 0]
No of clusters: 2
Number of samples in clusters: {1: 398, 0: 801}
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Thus, the graph contains two clusters. We conclude that there are 398 outliers (malicious dealers).